Pratical Work Grading Guidelines

(0) These grading guidelines are a work in progress and may undergo revisions.
(1) The practical work is evaluated as a demonstration of your ability to carry out (and communicate) scientific research. You are not graded on the "impact" of your results or whether they meet certain expectations but on how you produced, analyzed, and presented them. For example, you don't need to invent LSTM to get your degree; so finding out that an interesting hypothesis does not work is fine, if it has been executed properly.
(2) During the research and development phase, you are encouraged to seek and apply feedback from your supervisor as needed. However, for the report, you may receive high-level feedback from your supervisor only once, ensuring that the final written work is authentically your own.
(3) To receive an overall passing grade, you must achieve a passing grade in each of the three sections (Scientific Work, Scientific Documentation, and Report).

Not sufficient (5)	Sufficient (4)	Satisfactory (3)	Good (2)	Excellent (1)	Factor	Grade
Scientific Work					**0.50**	**0**
Research/technical skills					0.40	
*Inadequate knowledge of the research area. *Does not have a proper project management strategy (works sloppily). *Has acquired insufficient skills (e.g. coding) needed to fulfill the project.	*Limited knowledge of the research area, but lacks in-depth understanding and reasoning. *Must be regularly reminded of the importance of good project management strategy (work with precision), but does take this advice to heart. *Has acquired most of the necessary skills (e.g. coding) to conduct the project.	*Sufficient knowledge of the research area, and shows some in-depth understanding and reasoning. *Has a good project management strategy (works organized with precision) and, after an explanation, understands why certain procedures are necessary. *Has acquired the necessary skills (e.g. coding) to conduct good, reliable, and reproducible results.	*Good knowledge of the research area, shows an in-depth understanding of the conceptual (theoretical) framework. *Has a good project management strategy (works organized with precision) and understands why certain approaches are necessary, without a detailed explanation.	*Very good knowledge of the research area, shows an in-depth understanding of the conceptual (theoretical) framework and critically reflects on (new) information. *Has a professional project management strategy (works very well organized with high precision) and understands how certain approaches are connected and necessary, without any explanation.		
Scientific quality of the research process (1)					0.40	
*Only begins to understand the relevance of the research process and procedures after considerable explanation. *The resulting data and results are unreliable and unsubstantiated, lacking reproducibility and statistical significance, and are not of any use to the supervisor.	*Understands the relevance of research process and procedures after explanation, but is not yet capable of exploiting this to increase the quality and reliability of the conducted research. *The resulting data and results are only partly reliable and substantiated, with limited attention to reproducibility (e.g., absence of error bars) and no statistical significance testing.	*Understands the relevance of the research process and procedures without explanation and how experimental procedures can influence the reliability of data, but is not yet able to apply this knowledge to their project. *The resulting data and results are reliable, relatively well-substantiated, and include some basic elements of reproducibility (e.g., use of error bars) though limited statistical tests are applied.	*Understands the relevance of the research process and procedures without explanation and can pinpoint with some guidance which steps in the experimental procedure may lead to uncertainties, estimating the feasibility of the research. *The data and results are reliable, relatively well-substantiated, and include reproducibility measures (e.g., error bars) and some statistical significance tests, though with occasional guidance.	*Fully understands the research process and procedures without explanation, demonstrating a strong ability to estimate research feasibility and result reliability. *The data and results are reliable, very well-substantiated, and carefully interpreted within the context of related work. They include comprehensive reproducibility measures (e.g., error bars) and statistical significance testing, allowing for robust and reproducible conclusions.		
Motivation & Independance (2)					0.20	
*Struggles to work independently, relying heavily on supervisor guidance. *Doesn't seek feedback or is unresponsive to it.	*Requires firm guidance from the supervisor and demonstrates limited independence. *Rarely seeks feedback and shows minimal responsiveness to suggestions.	*Works independently with occasional support, creates plans, and generally adheres to time management. *Occasionally seeks feedback and makes some effort to apply it.	*Demonstrates strong independence, manages tasks effectively, and follows a well-organized plan. *Proactively seeks feedback, applies it, and updates the supervisor on progress.	*Exhibits full independence, strategically manages tasks and timelines, and adapts plans proactively as needed. *Consistently seeks feedback, thoughtfully applies it, updates the supervisor on progress, and constructively engages with criticism.		
Scientific Documentation					**0.20**	**0**
Code (only if applicable)					1.00	
*The code cannot be run or produces errors that prevent it from functioning. *The code lacks documentation, making it difficult or impossible to understand. *The code is poorly structured, with unclear naming and no modularity. *Data is not accessible, and no information is provided on where to find it. *Dependencies are not listed, and there are no instructions for setting up the environment.	*The code can be run but does not produce the same results as reported in the thesis. *Documentation is sparse, with unclear or incomplete comments and instructions. *The code is somewhat structured but has unclear naming and lacks modularity. *Data is not included in the repository and is not clearly indicated where to find it. *Dependencies are listed, but the instructions for setting up the environment are incomplete or unclear.	*The code can be run and produces similar results as reported in the thesis with minor discrepancies. *Documentation is adequate, with basic comments and a README file with general instructions. *The code is reasonably structured, with clear naming and some modularity. *Data is included in the repository or clearly indicated where to find it. *Dependencies are listed, and the instructions for setting up the environment are generally clear.	*The code can be run and produces the same results as reported in the thesis. *Documentation is thorough, with clear comments, a comprehensive README file, and detailed instructions for running the code. *The code is well-structured, with clear naming conventions and good modularity. *Data is included in the repository or clearly indicated where to find it, with precise instructions. *Dependencies are clearly listed, and the instructions for setting up the environment are detailed and easy to follow.	*The code can be run easily and produces the exact results as reported in the thesis, demonstrating full reproducibility. *Documentation is exceptional, with extensive comments, a detailed README file, and comprehensive instructions for running the code and reproducing the results. *The code is exceptionally well-structured, following best programming practices with clear naming conventions and high modularity. *Data is included in the repository or clearly indicated where to find it, with meticulous instructions and easy access. *All dependencies are meticulously listed, and the instructions for setting up the environment are thorough, ensuring a smooth setup process.		
Report (2-5k words (max. 10 pages) excl. references, story summary, and appendix; note that supervisors are *not* required to read the appendix)					**0.30**	**0**
Title					0.05	
*The title is missing or unclear, failing to convey the focus of the experiments.	*The title somewhat reflects the focus of the experiments but lacks clarity.	*The title adequately reflects the focus of the experiments.	*The title clearly and accurately reflects the focus of the experiments.	*The title is precise, descriptive, and effectively draws attention to the focus of the experiments.		
Introduction (min. 300 words)					0.10	
*The introduction does not provide background or context for the experiments. *The objectives of the experiments are not stated or are completely unclear. *There is no explanation of why the experiments are relevant or important.	*The introduction provides minimal background or context but is vague or incomplete. *The objectives of the experiments are mentioned but are not clearly defined. *There is little explanation of the relevance or importance of the experiments.	*The introduction provides sufficient background and context for understanding the experiments. *The objectives of the experiments are stated but could be more clearly defined. *There is a basic explanation of why the experiments are relevant or important.	*The introduction provides a clear and concise background, setting up the context for the experiments. *The objectives of the experiments are well-defined and easy to understand. *The relevance and importance of the experiments are well explained.	*The introduction is well-crafted, providing comprehensive background and context that leads seamlessly into the experiments. *The objectives of the experiments are clearly and thoroughly defined. *The relevance, importance, and potential impact of the experiments are compellingly explained.		
Methods (& experiments)					0.20	

*Too many/too few details are used. *No clear order has been used. *Important information is missing. Others are unable to repeat the research that is described in the report. *Experiments do not address the research question or highlight the contributions. *No mention of hardware or compute time is provided.	*Too many/too few details are used. *The order is not consistent. *Some important information is missing. Others may struggle to repeat the research that is described in the report. *Experiments are somewhat related to the research question but do not fully address it or highlight the contributions. *Hardware or compute time is briefly mentioned, but important details (such as type of hardware or total compute time) are missing or unclear.	*All techniques have been described, but sometimes too many/too few details are used. *A clear order is used and the sections are mostly concise and well-written. *At some points important information is missing, but others can repeat the research that is described in the report. *Experiments are generally in-line with the research question and highlight some contributions. *Basic information on hardware and compute time is mentioned, but more details are needed (e.g., only hardware type or only partial compute time is provided).	*All techniques have been described with enough detail, but the section should be adjusted before it can be used for publication. *A logical order is used and the sections are well described concisely. *All necessary information is provided for others to repeat the research that is described in the report. *Experiments effectively address the research question and highlight the contributions. *Both hardware and compute time are described in sufficient detail, but there may be minor gaps (e.g., total compute time is an estimate or hardware specifications lack specifics).	*Excellent description of the methods and experiments used, which can directly be used for scientific publication. *All sections are well described in a concise manner and logical order. *All necessary information is provided for others to repeat the research that is described in the report. *Experiments thoroughly address the research question and clearly highlight the contributions. *Detailed and comprehensive description of the hardware (e.g., GPU, CPU) and precise compute time, allowing full understanding of the resources needed to reproduce the experiments		
Results & their interpretation					0.20	
*Interpretation of the data is insufficient. *Results are poorly described and a coherent, understandable description of the data is missing. *Relation to the introduction, research aim or a broader scientific context is missing.	*Interpretation of the data is marginal. *Results are poorly described, and the story does not provide insight into why the research has been performed/how the data are related to the research aim. *Relation to the introduction and research aim is unclear and relation to a broader scientific context is lacking.	*The interpretation of the data and the quality of the figures and tables is generally good. *Results are described, but the story does not completely show why the research has been performed and how the data are related to the research aim. *The relation to the introduction and research aim is clear, but relation to a broader scientific context could be improved.	*Data are interpreted correctly. *Results are well described so that the audience understands why the research has been performed and how the data are related to the research aim. *Relation to the introduction, research aims, and a broader scientific context is clearly described.	*Data are all interpreted perfectly. *Results are very well described and the story is appealing to the reader. *Overall this section can be of interest for publication in a scientific journal. *Relation to the introduction, research aim and a broader scientific context is of high quality.		
Limitations & conclusion (min. 300 words)					0.10	
*Limitations are missing or insufficient. *Conclusion is missing or does not derive logically from the presented information.	*Limitations are poorly discussed. *Conclusion does not derive logically from the presented information.	*Some limitations of the work are discussed. *Conclusion is derived logically from the presented information.	*Major limitations are discussed. *Conclusion is strong and concise and derived logically from the presented information.	*All major limitations are discussed in-depth. *Conclusion is strong and concise and derived logically from the presented information.		
Referencing					0.05	
*References are largely missing or fail to support key statements. *Citation style is inconsistent or missing, making it difficult to identify sources.	*References are occasionally cited but are often unsubstantiated. *Citation style is used but inconsistently applied, with frequent errors.	*References generally support key statements, with only a few omissions. *Citation style is mostly consistent, with occasional formatting errors.	*References effectively support all statements, with minor exceptions. *Citation style is consistent, with only minor errors in formatting.	*References comprehensively support all statements, ensuring robust evidence throughout. *Citation style is fully consistent and professionally formatted throughout.		
Figures & tables (min. 4 figures/tables)					0.20	
*Data are presented sloppily and figure numbers and titles are not sufficient or even missing. *Information in the legends or graph axes is missing or inaccurate. *Error bars are missing or inaccurately applied.	*All data have been presented, but the quality of the figures and tables is often poor and no clear titles are used for figures and titles. *Legends or graph axes are incomplete or undefined. *Error bars are inconsistently used and may lack accuracy.	*All data have been well presented in a coherent manner and/or some titles or graph axes are not defined well enough. *Legends are present and fairly well described, but may be more concise. *Error bars are included but could be clearer or more consistent.	*The data is presented in a well-structured manner and figures and tables all include a sharp and concise title, graph axes are complete and clear. *Legends are present and well described containing all the information to understand the data that is presented. *Error bars are systematically shown.	*Data is presented in an excellent way. Figures and tables are of the highest quality. *Legends are all present and described in a concise, professional manner. *Error bars are systematically shown.		
Professional writing (grammar, spelling and tone; sentence structure and word choice; layout; equations)					0.10	
*Frequent grammar/spelling errors; lacks professional tone. *Convoluted, complex sentences that obscure meaning; lacks transition words. *Disorganized layout; information poorly structured with illogical paragraph flow. *Inconsistent or incorrect mathematical notation.	*Some grammar/spelling errors; lacks professional tone. *Some sentences are overly complex; lacks clarity and simplicity in word choice. *Organized layout but with minor issues in logical flow and paragraph structure. *Some inconsistencies in mathematical notation.	*Few grammar/spelling errors; mostly professional tone. *Clear sentences with concise, simple language; occasional issues with flow. *Well-organized layout with logical order; paragraphs are mostly coherent. *Mostly consistent mathematical notation, follows conventions.	*No grammar/spelling errors; professional tone. *Clear and concise sentences; good use of simple language with logical flow. *Well-organized layout; coherent and logically ordered paragraphs. *Consistent mathematical notation, follows conventions.	*No grammar/spelling errors; highly professional tone. *Clear, concise, and simple language; exceptional readability with logical flow. *Exceptional layout with well-structured, coherent, and logically ordered paragraphs. *Consistent, professional mathematical notation following conventions.		
				Proposed Grade (round up or down)		**0**